

Expanded Public Perception of Autonomous Cybernetic Avatars in 2025

Public Perception of Fully Autonomous Cybernetic Avatars in 2025: An In-Depth Analysis

Introduction

Fully autonomous cybernetic avatars (CAs)-robotic or digital agents capable of performing tasks and engaging in real-time interaction without ongoing human teleoperation-represent a transformative leap in human-machine symbiosis. Their increasing deployment across society in 2025 has been accompanied by intensive research, public demonstrations, heated media coverage, and extensive survey efforts exploring how people perceive, trust, and adapt to these entities. The expanding role of CAs in support, service, work, healthcare, and daily life has made the understanding of public acceptance and concerns particularly urgent.

This report provides an exhaustive, evidence-based analysis of contemporary public perceptions toward fully autonomous cybernetic avatars. It synthesizes large-scale survey data, controlled experiments, deployment case studies, and critical discourse from news, academic publications, and industry reports. Following a structured approach, each of the following sections targets a major dimension of public perception: trust, reliability, emotional realism, cost acceptance, societal integration, social and workplace contexts, healthcare, service sector feedback, public demonstrations, media coverage, comparison to semi-autonomous avatars, and ethical/legal implications.

Trust in Fully Autonomous Cybernetic Avatars

General Landscape and Empirical Data

Trust is both a precondition and a result of daily interactions with novel technology. Recent large public surveys and event-based studies, such as the 2025 Avatar Land demonstration in Osaka, reveal a robust but nuanced landscape of trust toward CAs. Among 2,285 participants, with 333 interacting with fully autonomous CAs, a notable **74.7% expressed positive attitudes** towards incorporating these avatars into their lives, and 39.3% indicated they were "very likely" to use such systems, with another 35.4% responding they would do so "if conditions are right"^[2]. Qualitative responses amplify these quantitative findings. Participants described high expectations particularly for "life-support and security robots (avatars)" in community safety and caregiving, indicating emergent trust in these applications. Crucially, trust hinges less on superficial anthropomorphism or visual design, and more on reliability and perceived utility-

though anthropomorphic cues do have a mediating impact when combined with empathy and contextual fit^{[4][6]}.

Table 1. Factors Influencing Trust in Fully Autonomous CAs

Trust Dimension	Survey/Study Findings	Example/Commentary
General Willingness	74.7% likely or conditionally positive (n=333, Avatar Land)	High receptivity but contingent on reliability
Emotional Connection	Stronger with culturally/ethnically matched or responsive avatars	Trust via perceived empathy ($\beta = 0.78$ mediation, $p < 0.05$)
Authentication Needs	Trust depends on robust avatar authentication systems	Called for in Expo 2025, academic panels
Privacy & Safety	Identity theft and impersonation cited as core trust issues	Emphasized in regulatory proposals, ethical panels
Contextual Trust	More trusted in security, service, caregiving tasks	Less trusted for untested/demonstration-only cases
Visual Realism	Not a determinant unless coupled with context-appropriate empathy	Visual cues secondary to functional reliability

Analysis:

Public trust in fully autonomous CAs is conditional and context-specific. Interest and even enthusiasm are apparent when avatars demonstrate concrete utility, especially in supportive or safety-critical roles. However, concerns persist around task reliability, clear identification/authentication (preventing misuse or impersonation), and data privacy. Academic studies confirm that trust is less a direct product of how human-like an avatar looks or acts, and more mediated through experiences of empathy, reliability, and transparency—underscored by the need for robust systems of avatar authentication and explainability^{[7][3]}.

Prominent public events and expert panels (e.g., Expo 2025 Osaka-Kansai) have further spotlighted requirements for institutional trust: standards for transparency, user notification protocols when an avatar is AI-driven rather than teleoperated, and ethical frameworks for trust reinforcement^[7].

Reliability Perceptions and Performance Feedback

Market Confidence and End-User Experience

Perceived reliability emerges as both the principal facilitator and the primary barrier to CA adoption. In the Avatar Land study, concerns over task completion and systemic robustness outweighed worry over costs or emotional realism, with 37.5% of those reluctant to use CAs citing "could not use well" as their rationale^[1]. This pattern holds across sectors—from workplace collaboration to healthcare and service domains.

Key statistical trends supporting reliability confidence:

- The AI-specific avatar market is growing explosively, reflecting business confidence in reliability and performance. Forecasts predict growth from \$0.8 billion in 2025 to nearly \$6 billion by 2032, or as high as \$118 billion globally by 2034, at compound annual rates above 30%.
- Controlled studies demonstrated intent recognition systems in CA-operated roles can achieve **classification accuracies of 90-96%**, effectively transforming operator input into polite, contextually appropriate avatar actions and shielding both end-users and operators from negative outcomes^[4].
- In workplace settings, avatars have been shown to **improve employee collaboration by 22%**, increase communication clarity (45% in hybrid teams), and boost feedback quality in performance reviews by 25%-all reliability-linked metrics^[8].

Table 2. Reliability-Linked Performance Metrics

Domain	Metric / Outcome	Study/Event
Task Execution (Avatar Land)	37.5% hesitancy due to reliability concerns	Avatar Land survey (2025)
Classification Accuracy	94-96% task/intent success in cashier/guardsman use-cases	Moral Support System Studies
Customer Satisfaction	81% user preference, higher politeness scores	Cashier scenario studies
Collaboration (Workplace)	22% collaboration, 17% team check-in, 41% brainstorming rise	PatentPC/Grandview surveys (2025)

Analysis:

Reliability perceptions remain the central axis around which trust and widespread adoption revolve. Even as technical and infrastructural advances (e.g., low-latency teleoperation, robust intent recognition, multi-agent collaboration) multiply, irregular failures or system breakdowns erode confidence and hinder acceptance. Academic and market data converge in recognizing the critical need for transparent error handling, context-appropriate fallback behaviors, and clear user guidance when CAs encounter limitations^{[3][4]}. Public demonstration failures (e.g., system falls in competitive scenarios) play a formative role in shaping user expectations and skepticism, highlighting that technical reliability and error management will remain the primary levers of acceptance for the foreseeable future^[9].

Emotional Realism and Human-Like Interaction

The Human Likeness Imperative

Emotional realism-the capacity for avatars to convincingly simulate or communicate emotions-constitutes a pivotal yet complex dimension of public perception. Interactivity studies and meta-analyses report that avatars engineered for human-like expressions, empathy, and cultural

adaptation are consistently rated as more trustworthy, relatable, and effective in fostering social bonds^[7].

- In experimental contexts, **anthropomorphic design significantly raised perceived empathy and trust**, positively affecting user experience (ANOVA: $F = 19.45$, $p < 0.001$; mediation β values up to 0.78)^[3].
- Facial expressivity, realistic eye-gaze, and nuance in gesture (e.g., avatars that blink or mirror cultural posture and gestures) have been shown to "add life" to avatars and be interpreted as intentional, contextually rich communication cues^[10].
- Ethnic and gender matching-the degree to which avatars correspond to the user's own identity-has been confirmed to enhance embodiment, emotional realism, and social attraction. Matching identity cues go beyond superficiality by lowering social friction and stimulating richer emotional responses^[7].

However, recent large-scale demonstrations indicate a shift:

While the public values emotional realism, **lack of it is not a primary barrier** to CA adoption when pitted against functional reliability. In the Avatar Land study, concerns about cost and human-like interaction were minimal compared to those tied directly to reliability and practical usage. Nonetheless, participant feedback still voiced a desire to "see how they will be used in the future" for improved communication, hinting at anticipated progress in emotional sophistication^[1].

Table 3. Emotional Realism - Drivers and Effects

Facet / Variable	Empirical Outcome	Reference
Expression & Gaze	"Adds life" and intentionality, boosts social believability	[3], [51]
Ethnic/Gender Matching	Enhanced embodiment/trust (effect sizes 0.22-0.26, Proteus effect)	[4], [28], [16]
Cultural Adaptation	Faster/powerful gestures preferred when matched to user culture	[16], [28]
Emotional Empathy	$\beta = 0.32-0.78$, strong UX mediation	[20], [3]

Analysis:

Emotional realism catalyzes trust and engagement, but in high-stakes or high-frequency deployments, users prioritize reliable function over subtle expressivity. Cultural adaptability and congruence with user identity magnify trust and acceptance where applied. The rapid evolution of deep learning, sentiment analysis, and affective computing now enables avatars to respond more believably and adaptively to human affect, though challenges remain in spontaneity and contextual appropriateness^[3].

Cost Acceptance and Economic Willingness

Economic Evaluation in a Transforming Market

Direct survey data from major demonstrations and cross-market research consistently indicate that **cost is not a dominant concern** limiting public adoption of CAs-so long as reliability and usefulness are adequately demonstrated. In the Avatar Land survey, only around 12.5% of respondents who were hesitant pointed to "seemed expensive" as a reason for disinterest^[2].

Market data substantiates soaring willingness-to-invest:

- The global digital avatar market reached **\$18-27 billion in 2023**, with forecasts predicting exponential growth to as much as **\$600 billion by 2030**, signifying high economic optimism and anticipated mainstreaming^[11].
- The U.S. market alone is forecast to climb from \$6.6 billion in 2023 to \$79 billion in 2030, at rates near or exceeding 45% CAGR.
- Enterprise adoption levels corroborate this willingness: 1 in 4 businesses have fully integrated avatars into workplace systems, and costs are increasingly justified by ROI in service, retail, and HR contexts^[12].

However, concerns remain around economic inequality and access:

Academic and policy reports urge caution: unless access to powerful CAs is widely available, substantial inequalities may be entrenched, particularly in domains like education, specialized healthcare, and telework. Calls for equitable dissemination, regulatory consideration, and public-access initiatives are recurrent themes in national and international discussions^[4].

Table 4. Cost Acceptance and Economic Willingness Metrics

Indicator	Value / Outcome	Implication
% users hindered by cost	~12.5% (surveyed)	Low relative barrier
Global market 2023→2030	\$18-27B → \$270-600B	Explosive growth, high willingness
US market 2023→2030	\$6.6B → \$79B	Strong regional investment
Enterprise CA adoption	25% (fully integrated systems)	Rising cost acceptance

Analysis:

The narrative around CA cost is fundamentally shifting, from skepticism or hesitancy (typical of other advanced technologies) to an enthusiastic willingness to pay for perceived value-provided reliability and demonstrable utility are clear^[1]. However, vigilance is warranted to prevent a societal divide, with potential for CA-driven opportunity gaps unless proactive measures for accessibility and affordability are enacted^[4].

Societal Role and Daily-Life Integration

Embedding Avatars into the Social Fabric

Public perceptions reveal a strong expectation that CAs will play integral roles in both **domestic life and professional spheres**. The Avatar Land demonstration found that almost **half of all respondents (47.8%) imagined uses of CAs in daily life**, with another third (32.5%) envisioning them as work partners or enhancers^[1]. Replicated household settings and real-world object manipulation underscored the avatars' potential for meaningful daily-life support, such as housekeeping, item retrieval, and assistance for the elderly or disabled.

Major public events and academic projects have framed CAs as pivotal in addressing some of the most pressing problems of modern societies:

- **Demographic crises:** CAs are posited as solutions to labor shortages connected to aging populations, especially in Japan and parts of Europe^[13].
- **Inclusion:** Avatars offer new opportunities for those with disabilities to work, socialize, and participate in culture-as witnessed at the Avatar Robot Café DAWN β, where remotely operated robots offer employment and interaction for people otherwise unable to participate in on-site work^[14].
- **Work-life balance:** Employees using avatars to maintain digital presence have reported improved boundaries and mental health, with reduced workplace anxiety and increased informal interactions resembling “watercooler talk”^[8].

Table 5. Societal Integration Use Cases

Context / Role	Findings	Examples
Domestic integration	47.8% in daily life, 32.5% in work (Avatar Land)	Housekeeping, child/elder/pet care, remote tutoring
Inclusion	Employment and social integration for disabled individuals	Café DAWN β, Butterfly Pavilion at Osaka Expo
Work/Life balance	Improved boundaries, 19% drop in anxiety (worker survey)	Avatar-enabled log-in/out for remote teams
Public safety / service	Robots as guards, clerks, customer service, community helpers	Public event demos, retail avatars

Analysis:

As avatars transition from spectacle to utility, their **societal role is normalizing**. The vision of “the human as a multi-bodied being”-proposed in Expo 2025 panels-captures this evolution. Yet, it is critical that policymakers, designers, and implementers remain alert to privacy, impersonation, and social cohesion risks, and ensure CAs are leveraged not just for efficiency but for genuine empowerment and inclusion^[14].

Social Contexts and Companion Use

Companionship, Empathy, and Social Bonds

Avatars are increasingly perceived not just as tools or assistants, but as **potential companions** in both digital and physical contexts. This is evidenced by:

- Large-scale studies reporting that avatars “offer more stable relationships for isolated or remote individuals,” amplifying social support in otherwise atomized conditions^[3].
- Companion-style avatars receive especially high acceptance when designed for psychological support or for use among the elderly, children, and those with disabilities-a trend corroborated by pilot projects and the popularity of virtual pets and emotionally responsive avatars^[15].
- The **anthropomorphization of CAs has led to striking real-world examples**, such as users confessing affection for AI assistants (e.g., Alexa), and the rise of “deathbots,” or avatars created to emulate deceased loved ones, which 14% of surveyed users found comforting despite ethical complexity.

Table 6. Companion and Social Context Acceptance

Interaction Aspect	User Perception / Behavior	Notable Findings
Emotional attachment	Stable long-term relationships, “love” for CAs	Alexa et al., deathbot comfort
Mental health support	Positive for stress, loneliness, therapy	50% of remote workers less isolated
Empathy-driven design	Perceived as more helpful, trustworthy	Embodiment via matching

Analysis:

Avatars’ capacity to fill social and emotional gaps is gaining both empirical support and public fascination. They are not only digital assistants but also **mirrors of human cognition** and potential “cognitive twins” capable of friendship-like bonds, particularly in situations of isolation, bereavement, or vulnerability^[7]. Care must be exercised, however, as growing attachment or reliance brings potential risks of psychological dependency and manipulation.

Workplace Collaboration and Telepresence

Revolution in Remote Work and Digital Presence

The workplace has been one of the earliest and most enthusiastic adopters of avatars-both semi- and fully autonomous. Empirical data and industry surveys paint a portrait of rising reliance, utility, and productivity:

- Avatars **increase employee collaboration by 22%, daily team check-ins by 17%, cross-functional collaboration by 28%**, and knowledge retention in training up to 32%. These

benefits are particularly pronounced among Gen Z and younger workers, who overwhelmingly prefer avatar-based interactions over webcams^[8].

- Leading enterprises and platforms (e.g., Microsoft Azure and Cloudforce) have incorporated avatars into onboarding, compliance training, internal Q&A, and even participatory meetings, pressing the boundaries between human presence and digital extension^[16].
- The **psychological safety fostered by avatars**-lowering appearance-based judgment, enabling anonymous feedback-has been recognized by more than half of HR leaders as vital to the future of employee experience^[8].

Nevertheless, the rapid scaling of avatars in the workplace has surfaced new challenges:

- **Authenticity, transparency, and accountability:** Concerns are emerging over how to distinguish between real participation and avatar-driven surrogate presence, emphasizing the need for clarity about when an employee is actively or passively represented.
- **Legal responsibility:** Questions of who is accountable for avatar statements or misbehavior-employee, employer, or vendor-are growing more complex as avatars become autonomous actors in organizational life^[7].

Analysis:

The digital workplace is inexorably shifting toward avatar-driven telepresence and hybrid forms of collaboration. As this technology matures, clear policies on avatar use, responsibility, and notification, along with ongoing training on best practices, will be critical to avoid erosions of trust or social cohesion^[7].

Healthcare Applications and Patient Views

The Empathy-Reliability Nexus in Medical Support

Healthcare, while less immediately associated with CAs than retail or education, is fast emerging as a transformative domain for avatar deployment. Several strands of evidence inform public and professional perceptions:

- AI-powered medical avatars exhibit strong acceptance among patients, particularly for mental health and chronic disease management. In survey studies, **patients showed a positive impact on trust and willingness to self-disclose sensitive information to avatars**, with perceived competence moderating these effects ($\beta = 0.74$, $p < 0.001$)^[15].
- Avatars serve practical roles in telemedicine, rehabilitation, and patient triaging. Mayo Clinic, Northwell Health, and HealthGauge employ avatars in symptom checking, therapy adherence, and continuous monitoring via wearable integration. These systems are credited with improving accessibility and reducing burdens on healthcare providers^{[18][19]}.
- Inclusive design enables avatars to be **used effectively by elderly, mobility-impaired, and linguistically diverse populations**, helping bridge access divides^{[7][14]}.

However, skepticism remains for high-stakes applications:

- Patient resistance correlates with concerns over avatar competence, privacy, and lack of cognitive flexibility compared to human clinicians-although, intriguingly, some patients trust avatars more than human physicians for certain conditions involving stigma or bias^[17].
- Regulatory frameworks, informed consent, and ongoing human oversight are called for as adoption scales^{[15][7]}.

Analysis:

CAs are poised to dramatically expand the reach and impact of healthcare-supporting, but not supplanting, human clinicians in critical functions. Further targeted research, ethical diligence, and transparent communication will continue to shape acceptance, especially as avatars progress beyond triage and scheduling to empathetic, context-aware patient engagement.

Service Sector Adoption and Customer Feedback

Service Efficiency, Personalization, and User Satisfaction

The **service sector** is among the earliest fields where avatars have demonstrated scalable value:

- In e-commerce and hospitality, **avatar deployment has resulted in tangible ROI**-for example, Chinese retailer Brother saw a 30% increase in revenue and \$2,500 earned in a two-hour avatar-driven livestream^[12].
- Holographic receptionists (e.g., RAVATAR) in hotels streamline guest check-in, reduce queues, and maintain high satisfaction while enabling post-pandemic contactless operations^[12].
- Survey participants and customers have repeatedly rated avatar-based interactions higher in politeness and effectiveness compared to non-avatar digital options, especially when support and intent recognition tools are present^[4].

Yet, as avatars assume greater visible roles, consumer expectations heighten:

- Customers expect **emotional realism and intuitive responses** but will show limited tolerance for errors or stilted behavior.
- Cultural adaptation and personalization are increasingly non-negotiable for high ratings in diverse, globalized markets^[20].

Analysis:

Customer service paradigms are being rewritten by avatar deployments, with clear gains in efficiency, satisfaction, and inclusivity. Organizations must prioritize continuous training, rapid personalization, and transparent fallback systems to maintain user trust and loyalty.

Public Demonstrations and Exhibition Case Studies

From Experimentation to Normalization

High-profile public exhibitions have played a decisive role in shaping both public perception and media narratives:

- **Avatar Land (Osaka, 2025):** Over 2,000 visitors, with 333 surveyed on fully autonomous CA interactions. Distinct scenarios (bedroom, kitchen, medical environments) underscored real-world potential and captured multidimensional user feedback^[1].
- **Expo 2025 Osaka-Kansai:** Robot avatars operated by online visitors, promoting barrier-free participation for remotely located or disabled individuals-regarded as a global first, with technological advances in wireless communication enabling over 100 avatars to operate in synchronized fashion^[21].
- **Avatar Robot Café DAWN β (Tokyo):** Blended teleoperated and autonomous roles for inclusive employment and interactive customer service, often cited in media and academic literature for its real-world social impact^[14].

Feedback from these cases underscores:

- Initial **awe and curiosity often give way to pragmatic judgments** about reliability, usefulness, and emotional engagement.
- Extended interaction prompts richer expectations: participants asked for more nuanced object handling, spontaneous dialogue, and contextual appropriateness in healthcare trial scenarios^[21].

Media Coverage and Public Discourse

Societal Narratives and Ethical Anxiety

Media coverage has largely tracked the arc from novelty to normalization, often amplifying key public concerns and amplifying fascination with avatar-enabled futures:

- Major outlets highlight tangible benefits-**empowerment for the disabled, improved work-life balance, and efficient customer interaction**-but also spotlight controversies, including issues of identity, impersonation, psychological dependency (e.g., “deathbots”), and the loss of human authenticity in the workplace^[6].
- Academic and technical conferences, such as IEEE ICRA's workshops on trust in autonomous systems, have generated high-profile, cross-sectoral discourse on demands for **safety, explainability, and accountability** for both users and bystanders^{[6][23]}.
- Cultural commentary increasingly touches on themes of **what it means to be human**, emotional continuity after death, and “avatars as central to cultural transformation,” sometimes blending philosophical speculation with practical critique^[6].

Public workshops, symposia, and citizen surveys: These participatory forums surface a wide array of hopes, ambitions, and critiques-reinforcing that public acceptance hinges as much on open dialogue and iterative design as on technical prowess^{[7][23]}.

Academic Research and Survey Data

Empirical Foundations and Theoretical Innovations

2025 has seen a maturation of **empirical research**, leveraging large samples and real-world demonstrations for robust insight:

- **Avatar Land:** 2,285 total respondents, 333 directly interacting with fully autonomous CAs; positive sentiment correlated to reliability, utility, and clear communication of autonomy status^[1].
 - **Dubai Service Sector Survey:** 1,000+ responses, strong overall acceptance, especially for physical/socially present avatars; preferences depended on context (appearance, setting, and function)^[20].
 - **Beijing Chatbot Avatar Study:** 160 participants; anthropomorphism and intelligence increased empathy and trust, but direct effects on user experience were mediated by perceived empathy rather than design alone^[3].
 - Cross-sectoral, international findings further emphasize **critical roles for transparency, identity verification, inclusive design, and adaptive feedback mechanisms**^[4].
-

Comparison of Semi-Autonomous and Fully Autonomous Avatars

Public Scrutiny and User Agency

Consensus in both empirical and theoretical literature confirms that **fully autonomous avatars draw greater scrutiny** and higher user expectations for spontaneous adaptation, reliability, and emotional realism than semi-autonomous systems requiring human oversight:

- Autonomously operating CAs, as demonstrated at Avatar Land and in competition scenarios, are assessed more stringently. Failures or “unnatural” behaviors are forgiven more readily in semi-autonomous modes where human intervention is apparent^[9].
 - Systems enabling seamless handoff between teleoperation and autonomy, and user notification protocols during AI-driven operation, are increasingly recommended as essential design features^[4].
 - Engagement is generally higher when agency is shared, as “subjective avatars” allow users to focus on high-value intent, while the system handles low-level adaptation and social cues. However, the more control is ceded to autonomy, the more carefully designers must manage expectations and articulate system boundaries^[3].
-

Ethical and Legal Implications

Regulation, Fairness, and Identity Management

Policy, ethics, and legal scholarship are scrambling to keep pace with the rapid deployment of CAs. Major themes include:

- **Certification and accountability:** Calls for international standards and robust authentication mechanisms (to prevent impersonation, identity theft, and misuse), as well as clear allocation of responsibility when avatars act independently in the world^{[6][23]}.
- **Privacy and data protection:** Stringent requirements are advocated for data security, especially in healthcare and education, where avatars process sensitive information^[15].
- **Algorithmic bias and inclusivity:** Fairness in design to prevent new forms of discrimination or exclusion, and explicit efforts toward accessibility across demographics^[7].
- **Intellectual property:** Navigating the rights to one's digital likeness, as illustrated in legal battles over the use of celebrity or deceased likenesses in avatars^[6].
- **Labor and workplace ethics:** Protecting worker agency, fair evaluation, and preventing "digital dilution" of human participation and accountability^[4].
- **Democracy and civic participation:** Proposals for using avatars to enhance inclusivity in political engagement (e.g., for hospitalized candidates or those with disabilities), juxtaposed against regulatory restrictions in areas like campaigning (defined as "billboards," or "puppets")^[4].

Table 7. Summary of Core Ethical-Legal Challenges

Challenge	Implication	Proposed Solution / Status
Identity/Authentication	Mitigate impersonation/privacy breaches	Certification frameworks
Accountability	Assign responsibility for autonomous action	Regulatory reform, insurance
Privacy	Protect sensitive user data in high-stakes domains	Data anonymization, encryption
Inclusion/Fairness	Avoid economic and algorithmic exclusion	Universal access initiatives
Algorithmic Bias	Ensure equitable outcomes across demographics	Transparency/auditability

Conclusion

The **pulse of public perception in 2025** reveals both substantial enthusiasm and clear, actionable concerns about the integration of fully autonomous cybernetic avatars into daily life. Trust is high where reliability is proven and empathy is built into interaction; reluctance surfaces

at points of failure, ambiguity, or exclusion. The trajectory of research, market trends, and real-world deployments unmistakably points toward further normalization and deepened integration-so long as technical advancement is matched by ethical design, regulatory clarity, and responsive user co-creation.

Practical acceptance will depend not only on solving engineering challenges, but also on designing with the broadest range of stakeholders in mind-anticipating challenges of identity, agency, fairness, and transparency. The avatar-symbiotic society is arriving, but building a digital future that genuinely enhances human agency and flourishing will require vigilance, humility, and ongoing interdisciplinary collaboration.

References (24)

1. *Public Evaluation on Potential Social Impacts of Fully Autonomous*
<https://arxiv.org/abs/2507.12741>
2. *Cybernetic Avatars and Society* . https://link.springer.com/chapter/10.1007/978-981-97-3752-9_9
3. *Frontiers* . <https://www.frontiersin.org/journals/computer-science/articles/10.3389/fcomp.2025.1531976/full>
4. *Public Evaluation on Potential Social Impacts of Fully Autonomous.*
<https://arxiv.org/html/2507.12741v1>
5. *Integration of the real & digital worlds* . <https://theme-weeks.expo2025.or.jp/en/program/detail/66e2ae1f9ffda.html>
6. *Cybernetic Avatars and Society.* <https://upcommons.upc.edu/bitstreams/2c34efe6-c616-4ac0-8e2f-171a132dd23c/download>
7. *Virtual Avatars in Workplaces: Adoption & Productivity Stats.* <https://patentpc.com/blog/virtual-avatars-in-workplaces-adoption-productivity-stats>
8. *A Cybernetic Avatar System to Embody Human Telepresence for*
<https://link.springer.com/article/10.1007/s12369-023-01096-9>
9. *ch14_Gillies_udpated2 - Goldsmiths, University of London.*
https://research.gold.ac.uk/id/eprint/2292/1/ch14_Gillies_udpated2.pdf
10. *What Is the Future of Personalized Avatars in 2025?*. <https://blog.powr.io/what-is-the-future-of-personalized-avatars>
11. *The Smart AI Receptionist: Your Solution for Seamless Customer Service.* <https://ravatar.com/ai-receptionist/>
12. *Goal 1: Overcoming limitations of body, brain, space and time.*
<https://www.jst.go.jp/moonshot/en/program/goal1/>
13. *Cybernetic avatars: Teleoperation technologies from in-body monitoring*
<https://www.science.org/doi/pdf/10.1126/scirobotics.adg1842>
14. *Transforming Patient Outcomes: The Role of AI Avatars.*

<https://insights.daffodilsw.com/blog/the-future-of-patient-engagement-how-ai-avatars-are-transforming-healthcare>

15. *AI Avatars: Redefining Human-Digital Interaction in the Enterprise Era.*

<https://techcommunity.microsoft.com/blog/azure-ai-foundry-blog/ai-avatars-redefining-human-digital-interaction-in-the-enterprise-era/4401973>

16. *Revolutionising patient care: the role of AI-generated avatars in*

https://academic.oup.com/eurheartj/article/45/Supplement_1/ehae666.3492/7839288

17. *A Whole-Body Integrated AVATAR System: Implementation of Telepresence*

<https://ramagazine.ieee.org/2025/06/24/a-whole-body-integrated-avatar-system-implementation-of-telepresence-with-intuitive-control-and-immersive-feedback/>

18. *Leveraging the Science to Understand Factors Influencing the ... - Springer.*

<https://link.springer.com/article/10.1007/s41347-022-00277-z>

19. *[2506.14268] Public Acceptance of Cybernetic Avatars in the service*

<https://arxiv.org/abs/2506.14268>

20. *Osaka-Kansai Expo: Robot Avatars to be Operated by Online Visitors*

<https://japannews.yomiuri.co.jp/science-nature/technology/20241102-220273/>

21. *Avatar Social Ethics Design.*

https://www.jst.go.jp/moonshot/en/program/goal1/appeal/files/11_ishiguro_ap08.pdf